Minibinder Design

Abstract

For SWARM, we wanted to develop a de novo minibinder targeting the VP3 capsid protein of deformed wing virus (DWV), with the ultimate goal of neutralizing the virus. However, VP3 presented two major challenges: much of the protein is buried within the viral capsid, while its exposed P-domain has few well-characterized binding hotspots suitable for small proteins. With no established minibinder-design strategy for this target, we developed our own computational pipeline combining structural analysis, multiple protein-design tools, benchmarking, and custom candidate filtering.

We analyzed 5,721 experimentally labeled protein designs across 41 targets to evaluate computational predictions and inform our filtering criteria. Using Mosaic, BindCraft, Boltz, and Protenix, we generated and evaluated minibinders through multiple design iterations. After an initial unsuccessful campaign, our refined approach produced a 79-amino-acid DWV-binding minibinder with an experimentally measured dissociation constant of approximately 604 nM. Our modeling played a central role in this result, demonstrating how computational analysis and iterative refinement can support protein design against difficult viral targets.

Introduction

Recent advances in computational protein design have introduced tools such as RFdiffusion, BindCraft, Mosaic, and Boltz, making it possible to generate thousands of potential minibinders against a protein target. However, identifying which computationally promising designs will successfully express and bind experimentally remains difficult.

For SWARM, we wanted to design a minibinder against DWV's VP3 capsid protein, a particularly challenging target for two reasons.

1. Limited Surface Accessibility

VP3 forms part of a larger viral capsid, meaning much of its structure is buried or constrained by neighboring capsid proteins. A region that appears favorable on an isolated VP3 structure may be inaccessible or structurally different within the assembled virus. This limits the number of binding surfaces available for an externally delivered minibinder.

2. Limited Suitable Binding Hotspots

Although VP3's protruding P-domain provides a potentially accessible region, identifying a suitable hotspot for a small minibinder is difficult. Its surface geometry and uncertainty in local

conformations complicate the prediction of favorable binding interfaces. Unlike larger antibodies, minibinders have relatively small interaction surfaces, making the selection of an appropriate epitope especially important.

The lack of established de novo minibinder designs against this target added another challenge. Rather than beginning with a previously validated minibinder-binding site, we had to investigate potential epitopes and determine which regions could realistically support a designed protein.

Furthermore, existing computational design tools use different approaches and evaluation metrics. A candidate with favorable predictions from one program may not perform similarly when evaluated by another, and high structural confidence does not necessarily translate into experimental binding.

We therefore developed our own computational design and filtering pipeline, combining multiple protein-design tools with structural analysis, benchmarking, and custom candidate-selection criteria. Our goal was to evaluate designs using more than a single confidence score and use the results to guide our experimental candidates.

Model Development

1. Structural Analysis and Epitope Selection

We began by analyzing the structure of VP3 to identify regions that could support minibinder binding. Our initial epitope investigation included BepiPred and DiscoTope, followed by additional analysis of solvent-accessible surface area (SASA), surface geometry, residue contacts, and hotspot accessibility.

We also compared the isolated VP3 P-domain structure (PDB 5G51) with the assembled DWV capsid (PDB 6F5J). This was particularly important because some regions appeared accessible on isolated or cropped structures but could not be verified in the full viral assembly.

Rather than relying on one predicted epitope, we investigated multiple regions across the P-domain.

Epitope Comparison

Region	Residues	Structural observations	Outcome
Initial C-terminal hotspot	I405, F408	Favorable isolated-domain geometry; residues unresolved in the analyzed virion structure	Round 1 produced no confirmed binders

EP-1	I302, Y359, R395	Alternative candidate region; crop-related exposure effects near Y359/R395 required caution	Computationally evaluated; no mapped binding hit
EP-2	V322, Q325, Y346	Alternative region including Q325, which is a predicted contact of the successful binder	Supported further Q325-region investigation
QP / A-ridge	Q325, P326	Both resolved and exposed in the assembly; 45.2 and 39.2 Å² SASA	Region targeted by confirmed binder
Alternative triad region	H277, S278, D294	Structurally accessible in part, but proposed catalytic rationale was not supported by later evidence	Four submitted designs; no confirmed binding

Our initial campaign targeted the C-terminal region surrounding Ile405 and Phe408. Although this site appeared promising in the isolated P-domain structure, subsequent analysis revealed that these residues were unresolved in the corresponding region of the assembled capsid structure.

We also identified artifacts introduced by cropping the target. For example, Val384 had approximately 79 Å² of solvent-accessible area in the assembly compared with 130 Å² in a cropped structure, creating an artificially favorable surface for computational design.

These results highlighted the importance of evaluating potential hotspots in their biological structural context rather than relying exclusively on isolated protein structures.

We expanded our analysis to alternative epitopes, including regions around Gln325 and Pro326. Both residues remained accessible in our assembled-capsid analysis, making them suitable candidates for further computational investigation.

We also incorporated hotspot analysis from PRINCE, our independently developed epitope-selection software, to evaluate alternative binding regions. Details of PRINCE's development and benchmarking are provided in our Software section.

Interactive Figure 1: VP3 Epitope Explorer

[Insert interactive molecular viewer]

The viewer will show the 5G51 P-domain and the corresponding region in the 6F5J capsid. Users will be able to select different candidate epitopes, highlight their residues, and compare their accessibility and predicted interactions.

Each selection will display the hotspot residues, associated design campaign, structural observations, and experimental outcome where available. The final QP region will also display our experimentally confirmed minibinder in its predicted binding pose.

This will allow judges to directly examine the different regions we investigated and understand how structural modeling informed our design process.

Researcher Feedback

Throughout this process, we also consulted researchers working in protein design and DWV biology.

Tudor Coțet at Adaptyv provided feedback relevant to our computational design and experimental screening process. Benjamin Lamp provided biological context supporting the investigation of the DWV P-domain, while Tatsuhiko Kadowaki encouraged us to explore multiple binding regions rather than restricting our designs to a single site.

These discussions complemented our structural analysis and helped inform the direction of subsequent design campaigns.

2. Benchmarking Protein-Design Methods

One of the main challenges in computational minibinder design is determining which predictions actually matter when selecting candidates for experimental testing.

Different programs generate distinct structural-confidence scores, but these metrics do not necessarily indicate whether a protein will fold, express, or bind successfully.

To investigate this, we assembled a dataset of 5,721 experimentally labeled protein designs across 41 targets from Proteinbase, containing 849 binders and 4,872 non-binders.

We analyzed structural-confidence metrics, predicted interface properties, and sequence characteristics to understand their relationship with experimental outcomes.

Computational metric	Pooled ROC-AUC
Minimum ipSAE	0.806

ipSAE	0.801
Boltz-2 ipTM	0.628

Minimum ipSAE performed substantially better than ipTM in the pooled comparison. However, its performance decreased when we evaluated designs targeting the same protein, showing that pooled metrics can overestimate their usefulness for selecting binders against a specific target.

We also investigated whether combining a large number of metrics improved predictive performance. A naive ensemble of 38 metrics produced an ROC-AUC of only 0.541, substantially below the strongest individual metrics.

These results informed our filtering strategy. Rather than combining every available score, we focused on selecting useful computational metrics and evaluating them alongside structural and sequence-based evidence.

3. Developing Our Filtering Framework

We developed a multi-stage filtering framework to identify promising candidates from the large number of computationally generated designs.

Our goal was not simply to select proteins with the highest predicted binding confidence, but to evaluate whether they formed plausible complexes, maintained suitable structural properties, and could reasonably be prioritized for experimental testing.

Structural Filtering

We evaluated predicted interface confidence, protein folding, intermolecular contacts, steric compatibility, and agreement between structural models.

For difficult targets such as VP3, we also considered whether the predicted interface remained accessible in the assembled capsid.

Sequence Filtering

We incorporated sequence-based checks addressing cysteine content, glycine and glutamate fractions, hydrophobicity, aggregation-related properties, and sequence entropy.

These criteria helped evaluate properties that were not captured by protein-complex confidence scores alone.

Repeated Structural Predictions

We found that predictions for the same design could vary significantly between runs. One candidate produced minimum ipSAE values ranging from 0.000 to 0.676, demonstrating how selecting only the highest-scoring prediction could be misleading.

We therefore considered repeat predictions and their consistency when evaluating promising candidates.

Manual Structural Evaluation

Finally, we manually inspected predicted complexes to identify questionable interfaces, unusual structural features, and potential issues that automated metrics might overlook.

Combining these approaches allowed us to develop a more systematic candidate-selection process.

4. Integrating Multiple Design Tools

Our pipeline combined several protein-design methods, including Mosaic, BindCraft, and BoltzProt, with structural evaluation using Boltz and Protenix.

Rather than relying on one generator, we investigated whether different computational methods could explore distinct binding regions and protein structures.

Across 1,176 pairwise design comparisons, BindCraft and Mosaic candidates exhibited a median interface Jaccard similarity of 0.000, with a median distance of approximately 42.9 Å between their predicted interface centers.

This demonstrated that the tools explored substantially different binding regions in our candidate pool, supporting the use of multiple generators rather than concentrating exclusively on one method.

We then combined their outputs and applied source-specific filtering, structural evaluation, and sequence-diversity constraints.

For our second campaign, we assembled a computational pool of 103 candidate entries, from which 20 designs were ultimately submitted for experimental testing.

Interactive Figure 2: Candidate Filtering Pipeline

[Insert interactive candidate-filtering visualization]

This visualization will show how generated candidates moved through structural evaluation, sequence filtering, cross-model comparison, and diversity selection. Users will be able to inspect representative candidates and the criteria used to evaluate them.

Iterative Modeling and Results

1. Initial Design Campaign

Our first design campaign consisted of 15 submitted minibinders targeting the VP3 P-domain.

Several candidates received favorable computational predictions, including one with an ipSAE of 0.927 and an ipTM of 0.958.

Despite these predictions, none produced a confirmed experimental binding hit. Nine received negative binding calls, while six lacked sufficient assay loading for definitive assessment.

We investigated potential causes of failure, including hotspot selection, sequence composition, and the limitations of complex-level structural confidence.

One finding was that our initial designs contained unusually high proportions of alanine, glutamate, lysine, and arginine, averaging approximately 48% of residues.

We traced this composition bias to restrictive sequence-generation settings and subsequently reduced the MPNN weight from 8.0 to 4.0 while increasing sampling temperature from 0.01 to 0.1.

We also expanded our evaluation to include independent monomer-folding checks rather than relying exclusively on predicted protein-complex confidence.

2. Refined Design Campaign

Our second campaign incorporated additional hotspot evaluation, improved sequence generation, structural filtering, and greater diversity between selected candidates.

We evaluated designs against multiple VP3 binding regions and considered their structural consistency across computational models.

One selected design, l79_s469532_mpnn9, was a 79-amino-acid BindCraft minibinder targeting the Gln325/Pro326 region.

Its predicted complex showed strong structural confidence:

Metric	Value
Protenix ipTM	0.949
Boltz-2 ipTM	0.936

Minimum ipSAE	0.838
Monomer pLDDT	0.896
Self-consistency RMSD	1.826 Å

Importantly, this candidate was not the highest-ranked design by minimum ipSAE alone. Another candidate achieved a higher score without confirmed experimental binding.

This reinforced our benchmark finding that individual structural-confidence scores are insufficient for reliably prioritizing experimental binders.

3. Experimental Validation

Our second campaign produced one confirmed DWV-binding minibinder among 20 submitted designs.

Biolayer interferometry (BLI) measurements across two replicates produced a combined dissociation constant of:

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K_D \approx 604\ \mathrm{nM}
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This confirmed moderate-affinity binding to the experimentally tested DWV target construct.

Across both campaigns, we submitted 35 minibinder designs, with one confirmed binding hit.

Although viral neutralization has not yet been established, this result demonstrated that our modeling pipeline successfully supported the development of an experimentally functional binder.

Submitted Minibinder Results

The complete wiki table will include all 35 submitted designs, their design campaigns, and the available experimental binding outcomes.

Campaign	Submitted	Confirmed binders	Negative binding calls	No usable binding call

Round 1	15	0	9	6
Round 2	20	1	9	10
Total	35	1	18	16

Interactive Figure 3: Experimental Candidate Explorer

[Insert sortable table of all 35 minibinders]

The interactive table will display individual design IDs, generation methods, predicted epitopes, selected computational scores, and experimental outcomes.

Users will be able to filter by campaign or experimental result, compare computationally promising candidates with experimental failures, and inspect the confirmed binder in more detail.

Implications for SWARM

Our computational modeling played a central role in SWARM's development of a DWV-binding minibinder. The structural challenges of VP3 required us to investigate potential binding regions carefully and develop a systematic approach for generating and evaluating candidates.

Our initial campaign demonstrated that favorable computational confidence scores did not guarantee experimental binding. By incorporating structural analysis, benchmarking, sequence filtering, and additional evaluation methods, we refined our design strategy and ultimately obtained an experimentally confirmed binding hit.

Our modeling also revealed important limitations in current computational protein design, particularly the importance of biological assembly context, sequence properties, and reproducibility across structural predictions.

These findings provide a foundation for future minibinder development against VP3 and other challenging protein targets.

Conclusion

Through structural analysis, computational benchmarking, and iterative protein design, we developed a modeling pipeline for generating and evaluating minibinders against DWV VP3.

Our approach integrated multiple protein-design tools with custom filtering and experimental feedback, ultimately producing a 79-amino-acid minibinder with confirmed binding at approximately 604 nM.

Our modeling was instrumental in SWARM's successful minibinder development and established a more systematic framework for evaluating difficult de novo protein-design targets.